

Global Superstore DAX Measures Documentation

Calculated Metrics Reference

This document lists the DAX measures created for the Global Superstore Power BI dashboard, the logic behind each, and the business purpose each measure serves. All measures are built on the Orders table (and a Date table for time-intelligence measures) using standard star-schema relationships.

Summary Table

Measure	Formula	Purpose
Total Sales	SUM(Sales)	Calculates total revenue across all orders.
Total Profit	SUM(Profit)	Calculates total profit across all orders.
Total Orders	DISTINCTCOUNT(Order ID)	Counts the number of unique orders, avoiding duplicate line-item counts.
Total Customers	DISTINCTCOUNT(Customer Name)	Counts the number of unique customers in the current filter context.
Profit Margin	Total Profit ÷ Total Sales	Measures overall profitability as a percentage of revenue.
Average Sales per Order	Total Sales ÷ Total Orders	Measures the average value of a single order.
Average Profit per Order	Total Profit ÷ Total Orders	Measures the average profit generated per order.
Average Discount	AVERAGE(Discount)	Measures the average discount rate applied across orders.
Loss-Making Orders	COUNTROWS(FILTER(Orders, Profit < 0))	Counts the number of orders that generated a loss, highlighting profitability risk.
Sales Growth %	(Current Period Sales – Prior Period Sales) ÷ Prior Period Sales	Measures the percentage change in sales versus the prior year, for trend and growth analysis.
Profit Growth %	(Current Period Profit – Prior Period Profit) ÷ Prior Period Profit	Measures the percentage change in profit versus the prior year, for trend and growth analysis.

Full DAX Syntax

The exact DAX code for each measure is listed below for reference and reuse.

Total Sales

Total Sales = SUM(Orders[Sales])

Total Profit

Total Profit = SUM(Orders[Profit])

Total Orders

```
Total Orders = DISTINCTCOUNT(Orders[Order ID])
```

Total Customers

```
Total Customers = DISTINCTCOUNT(Orders[Customer Name])
```

Profit Margin

```
Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
```

Average Sales per Order

```
Average Sales per Order = DIVIDE([Total Sales], [Total Orders], 0)
```

Average Profit per Order

```
Average Profit per Order = DIVIDE([Total Profit], [Total Orders], 0)
```

Average Discount

```
Average Discount = AVERAGE(Orders[Discount])
```

Loss-Making Orders

```
Loss-Making Orders = COUNTROWS(FILTER(Orders, Orders[Profit] < 0))
```

Sales Growth %

```
Prior Year Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date])) Sales  
Growth % = DIVIDE([Total Sales] - [Prior Year Sales], [Prior Year Sales], 0)
```

Profit Growth %

```
Prior Year Profit = CALCULATE([Total Profit], SAMEPERIODLASTYEAR('Date'[Date])) Profit  
Growth % = DIVIDE([Total Profit] - [Prior Year Profit], [Prior Year Profit], 0)
```

Notes

- Total Orders and Total Customers use DISTINCTCOUNT rather than COUNTROWS because the raw data is at line-item level, and a single order or customer can appear across multiple rows.
- All ratio measures (Profit Margin, Average Sales per Order, Average Profit per Order, growth measures) use DIVIDE() rather than the "/" operator so they return 0 instead of an error when the denominator is blank or zero.
- Sales Growth % and Profit Growth % rely on a dedicated Date table marked as the model's date table, using SAMEPERIODLASTYEAR() to compare like-for-like periods.
- Loss-Making Orders supports the loss-making product investigation in the Advanced Data Analysis document and can be sliced by Category, Sub-Category, or Region to isolate where losses are concentrated.